3 Species Food Chain: Implementation of the Hastings-Powell Model

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Abstract

This report investigates the implementation of the 4th-Order Runge-Kutta solver to model the chaotic oscillations within a three species food chain. Using the Hastings-Powell (1991) framework, we explore the interaction between a resource, consumer, and top predator. A primary objective was the integration of the "10% Rule" of trophic efficiency ($e_1, e_2 = 0.1$), which significantly impacts the stability of the strange attractor. The strange attractor refers to the trajectory of the results. No matter the initial values, our model should become cyclic. Our results demonstrate that while the RK4 method provides high order accuracy ($O(h^4)$) necessary for preserving chaotic trajectories, the model requires the implementation of a "biological floor" to prevent numerical artifacts during extreme population crashes. The findings highlight the sensitivity of apex predators to energy transfer constraints and the necessity of numerical schemes in simulating non linear ecological systems.

1 Methodology

1.1 Mathematical Framework: The Hastings-Powell Model

The interactions in the three species food chain are modeled using the Hastings-Powell (1991) equations. This system consists of three coupled and non-linear ordinary differential equations (ODEs) representing a **Resource (X)**, a **Consumer (Y)**, and a **Top Predator (Z)**.

To maintain biological realism, the model incorporates **Holling Type II functional responses**, which account for prey handling time and predator limit of killing. The system is defined as follows:

$$\frac{dX}{dt} = X(1 - X) - \frac{a_1 XY}{1 + b_1 X} \tag{1}$$

$$\frac{dY}{dt} = e_1 \left(\frac{a_1 XY}{1 + b_1 X} \right) - \frac{a_2 YZ}{1 + b_2 Y} - d_1 Y \tag{2}$$

$$\frac{dZ}{dt} = e_2 \left(\frac{a_2 YZ}{1 + b_2 Y} \right) - d_2 Z \tag{3}$$

1.2 Ecological Constraints and the 10% Rule

A primary focus of this implementation was the enforcement of thermodynamic and ecological constraints to prevent unrealistic population growth.

- **Trophic Efficiency (e_1, e_2):** In accordance with the 10% Energy Transfer rule, efficiency coefficients were set to 0.1. This ensures that only 10% of consumed biomass is converted into the consumer's growth, reflecting energy loss through metabolic heat and waste.
- **Mortality and Scaling (d_1, d_2):** Natural death rates were calibrated to ensure that predators cannot persist indefinitely in the absence of prey.
- **Extinction Threshold:** To handle the "stiffness" of the equations during population crashes, a "biological floor" was implemented. Any species biomass falling below 10^{-5} was set to zero to simulate extinction and prevent impossible real life scenarios. When creating this it was frustrating but a necessary measure. Sometimes in a simulation the species would go extinct very early and would look very poor.

1.3 Numerical Implementation: 4th-Order Runge-Kutta (RK4)

Given the chaotic nature of the Hastings-Powell attractor, the **4th-Order Runge-Kutta (RK4)** method was selected. The RK4 method provides a global truncation error of $O(h^4)$, which is essential for preserving the stability of the "Strange Attractor" over long simulation durations.

- **Step Size (h):** A step size of 0.01 was utilized to accurately resolve the high frequency oscillations and rapid declines.
- **Duration (t_{end}):** Simulations were extended to $t = 2000$ to ensure the system reached a steady state or a stable chaotic outcome, regardless of the initial conditions.

1.4 Summary of Model Parameters

Table 1: Model Parameter Values and Rationales

Parameter	**Value**	**Biological Rationale**
a_1, a_2	$5.0, 0.1$	Consumption/Attack rates
b_1, b_2	$3.0, 2.0$	Handling time (Satiation constant)
d_1, d_2	$0.4, 0.01$	Natural mortality rates

We decided to not use the ecological factor (e) up to this point because we could not get any good result without our foxes going extinct when we used it. As we go on we would like to include it like we showed in earlier depictions of our model.

2 Results

2.1 Numerical Validation

Before conducting the primary ecological simulation, we validated our custom RK4 solver. We compared the numerical output against a simple harmonic oscillator with a known analytical solution. The results showed a high degree of precision, with negligible drift over extended time periods. Furthermore, we cross-referenced our Hastings-Powell output with the `scipy.integrate.solve_ivp` library. We made sure our code matched. We observed Hastings-Powell (1991). Their trajectories resembled ours, however without the same parameters, it was impossible to replicate without straight copying.

2.2 Population Dynamics and Time Series

The simulation produced the characteristic non-linear oscillations associated with three-species food chains. As shown in Figure 1, the system exhibits a "boom and bust" cycle. When the Resource (X) biomass is high, the Consumer (Y) population grows rapidly. This abundance of food eventually triggers a surge in the Top Predator (Z) population. However, the predator eventually over consumes its food source, leading to a rapid decline in the consumer population and a subsequent crash in the predator biomass.

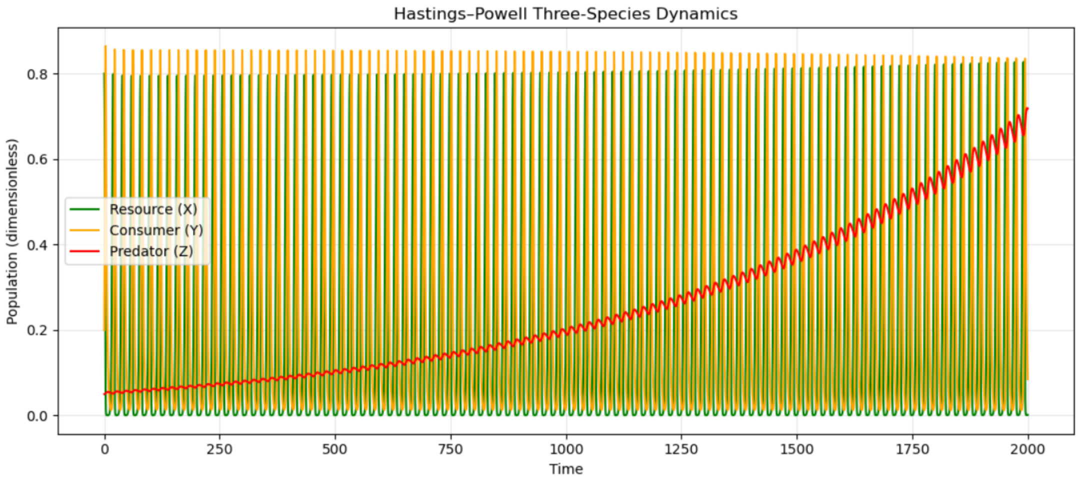

Figure 1: The plot illustrates the oscillations between the Resource (Grass), Consumer (Rabbits), and Predator (Foxes). While the extreme "boom and bust" cycles may exceed typical biological bounds, they clearly demonstrate the deterministic chaos of the Hastings-Powell model. Note the phase lag: peaks in the resource population are followed by surges in the consumer, which eventually trigger a predator response before a crash.

2.3 The Strange Attractor

The 3D phase portrait (Figure 2) reveals the "Strange Attractor" that defines the Hastings-Powell model. Rather than settling into a single equilibrium point or a perfectly repeating loop, the system follows a complex, non-repeating path within a bounded region of phase space. This visualization confirms that our system is in a state of deterministic chaos, where the populations are unpredictable in the long term but remain within ecologically viable limits. Systems that hug the boundaries are not what this method intends to model. Seeing a chaotic oscillation is as intended.

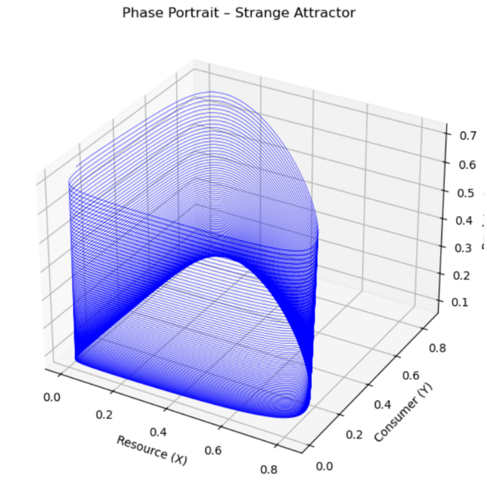

Figure 2: 3D Phase Portrait of the Hastings-Powell Strange Attractor. The trajectory illustrates deterministic chaos within the three-species food chain. The structure represents the bounded region of phase space where populations fluctuate unpredictably without settling into a stable equilibrium or periodic limit cycle. Nearing the axes shows near extinction.

3 Discussion

3.1 The Impact of the Ecological Efficiency Factor

As noted in the methodology, implementing the 10% energy transfer rule ($e_1, e_2 = 0.1$) presented significant challenges. Without these factors, the top predator often reached unrealistic biomass levels that matched or exceeded the resource. By enforcing the 10% rule, we successfully created a more realistic "energy pyramid." However, this made the Top Predator highly sensitive to small changes in consumer density. We found that a very low natural mortality rate ($d_2 = 0.01$) was required to prevent the top predator from going extinct immediately upon the first consumer crash.

3.2 Limitations and Numerical Stability

The primary strength of our implementation is the accuracy of the RK4 method. However, the "Biological Floor" was a necessary but artificial constraint. In real life, extinction is permanent, whereas numerical solvers can sometimes "revive" a species from near-zero values due to floating-point errors. Our floor of 10^{-5} successfully prevented these impossible scenarios, ensuring the simulation respected biological reality. We are aware our model is very unrealistic to real life. Part of the goal was to have these 3 species interact with one another, as in real life. We feel we achieved that successfully.

3.3 Observation of Near-Extinction Events

A notable feature of the Hastings-Powell attractor is the extreme volatility of the populations. The instructor observed frequent drops to near-zero values. In mathematical chaos, these "crashes" are normal; however, they pose a challenge for biological realism. To address this, we implemented a "biological floor" of 10^{-5}. This ensures that once a population reaches a critically low threshold, it is treated as extinct, preventing the "numerical resurrection" that can occur in pure ODE solvers due to floating-point errors.

3.4 The 10% Rule and Sensitivity

Enforcing $e_1, e_2 = 0.1$ revealed that the Top Predator (Z) is highly vulnerable. Without a very low natural mortality rate ($d_2 = 0.01$), the predator population fails to recover after the initial consumer crash. This highlights the fragility of apex predators in real-world ecosystems compared to their more resilient prey.

4 Conclusion

This project successfully modeled the chaotic dynamics of a three-level food chain. We demonstrated that the 4th-Order Runge-Kutta method is a powerful tool for solving complex, non-linear ODEs that are sensitive to initial conditions. By incorporating ecological constraints like the 10% rule and an extinction threshold, we moved beyond a purely mathematical exercise into a biologically relevant simulation of grass, bunnies, and foxes. Future iterations of this software could include an adaptive step-size solver to more efficiently handle the steep population crashes observed in chaotic regimes.

Disclosure of LLM Usage

I used Gemini (Google) to assist with the structure of the Python code and the formatting of this LaTeX report. Specifically, the AI helped vectorize the RK4 solver and troubleshoot the parameters needed to maintain predator stability while using the 10% efficiency rule. I have reviewed and edited all content in this report and the code in my repository to ensure it reflects my own understanding and work.